

Survey Paper on Medical Cost Prediction Using Machine Learning

GTU Data Science Mini Project

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Abstract

Healthcare expenditure is one of the fastest-growing economic burdens worldwide. Insurance companies require reliable methods to estimate annual medical charges for policy pricing and risk management. This survey paper reviews existing approaches for health insurance cost prediction and presents a proposed machine learning solution based on the Medical Cost Personal Dataset. The proposed system uses exploratory data analysis, feature encoding, and a Random Forest Regressor to predict charges from demographic and lifestyle attributes. Experimental results on a held-out test set achieve MAE of approximately \$1,982, RMSE of \$2,517, and R^2 of 0.93, demonstrating strong predictive performance suitable for practical decision support.

1. Introduction

Medical insurance pricing depends on accurate estimation of expected healthcare costs for individual beneficiaries. Traditional actuarial methods rely on manual rules and historical averages, which may not capture complex interactions between age, body mass index (BMI), smoking habits, family size, and geographic region.

With the availability of structured health insurance datasets and advances in supervised learning, data-driven models can learn non-linear patterns from historical records. The objective of this mini project is to design an end-to-end predictive system that estimates annual insurance charges (in USD) and supports both batch analysis and interactive prediction through a web application.

The problem is formulated as supervised regression: given features age, sex, BMI, number of children, smoker status, and region, the model outputs a continuous charge value. This survey paper summarizes prior work, compares existing techniques, and describes the methodology adopted in the present implementation.

2. Existing Methods (Literature Survey)

Researchers have applied several machine learning and statistical techniques to predict medical or insurance costs. The following approaches are commonly reported in the literature:

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2.2 Decision Tree and Ensemble Methods

2.3 Support Vector and Neural Network Approaches

2.4 Comparative Observations

- BMI and age show positive correlation with charges in exploratory studies.
- Region effects are smaller but statistically visible in aggregate analysis.
- Ensemble tree models consistently rank among the best performers for this problem size.

3. Proposed Approach

The proposed system follows a standard data science pipeline aligned with industry practice and GTU mini-project guidelines. It uses the Medical Cost Personal Dataset containing 1,338 records with seven attributes: age, sex, BMI, children, smoker, region, and charges (target).

3.1 Data Preprocessing

- Separate features (X) and target (y = charges).
- Encode categorical variables (sex, smoker, region) using One-Hot Encoding.
- Split data into 80% training and 20% testing with fixed random seed.

3.2 Exploratory Data Analysis

- Box plot: smoker vs charges (large premium gap for smokers).
- Bar chart: average charges by region.

3.3 Model Selection and Training

3.4 Evaluation Metrics

- Mean Ab

- Root Mean Squared Error (RMSE): penalizes large errors.
- R-squared (R^2): proportion of variance explained by the model.

4. Implementation and Results

The project is implemented in Python using pandas, NumPy, scikit-learn, matplotlib, seaborn, and Jupyter Notebook. A Streamlit application provides an industry-style deployment interface. On the test set, the proposed Random Forest model achieved:

Metric	Value
Mean Absolute Error (MAE)	\$1,982.25
Root Mean Squared Error (RMSE)	\$2,517.21
R-squared (R2)	0.9331

Sample prediction: a 39-year-old female with BMI 29.4, two children, non-smoker, northeast region yields an estimated annual charge of approximately \$14,623. Bulk prediction for 300 profiles is exported to CSV for operational demonstration.

5. Advantages and Limitations

Advantages

- Handles categorical and numeric inputs in a unified pipeline.
- Reproducible notebook workflow and deployable web interface.

- High accuracy

Limitations

- Model does not include clinical diagnosis or hospital claim details.
- Fairness across demographic groups requires separate ethical review.

- Dataset bias

6. Conclusion

This survey paper reviewed traditional and machine learning methods for medical insurance cost prediction and presented a Random Forest-based solution with strong empirical results. The proposed approach satisfies mini-project requirements: clear problem definition, literature context, reproducible methodology, measurable output, and practical deployment via Streamlit. Future work may include hyperparameter optimization, XGBoost comparison, and larger claim databases.

7. References

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